

Deep Learning-Based Secure Real Time Face Recognition System

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Abstract: Rapid digital transformation has precipitated widespread automation of operational processes across both domestic and organizational contexts. Accurate attendance management constitutes a critical administrative function within educational institutions and enterprise environments alike. Conventional attendance monitoring systems are, however, susceptible to significant security vulnerabilities, including identity spoofing and fraudulent record manipulation. Facial morphology represents a prominent biometric modality owing to its non-intrusive acquisition properties and its foundational role in human identity verification. This study presents the design and implementation of a deep learning-based automated attendance system capable of detecting and logging the presence of employees or students in real time. The proposed system is implemented in Python and employs a Convolutional Neural Network (CNN) architecture to extract discriminative facial features and perform individual identification. The Viola–Jones Haar-cascade detector is applied for real-time face localisation within video streams. Anti-spoofing security is enforced through a Chaotic Logistic Key Mapping Algorithm for facial data encryption, integrated within a graphical user interface developed using the PyQt framework. User enrolment records and attendance logs are persisted in a PostgreSQL relational database. The synergistic integration of the CNN-based recognition module and the Viola–Jones Haar-cascade detector yields measurably improved face recognition accuracy compared with either algorithm in isolation. Experimental results demonstrate the system's effectiveness across three critical dimensions: face recognition accuracy, real-time face localisation, and resilience against spoofing attacks, thereby establishing a robust and comprehensive biometric attendance solution. The proposed system benchmarks four face recognition algorithms—Eigenfaces, Local Binary Pattern Histograms (LBPH), Support Vector Machine (SVM) classification, and CNN—and reports their performance across standard evaluation metrics. Algorithm performance is evaluated and compared using four standard classification metrics: accuracy, precision, recall, and F1-score. Among the evaluated algorithms, the CNN model achieves superior face recognition accuracy and demonstrates consistently strong performance across all other reported evaluation metrics. The CNN model was evaluated across four train-to-test partition ratios—60:40, 70:30, 80:20, and 90:10—with the 90:10 split yielding the highest face recognition accuracy.

Keywords: Facial Recognition, Viola Jones, Convolutional Neural Network, Chaotic Logistic Mapping, PyQt.

1 Introduction

Human faces serve as a fundamental modality for individual identification across a wide range of perceptual and computational contexts. Face recognition is a biometric identification method that extracts and encodes a subject's discriminative facial features to enable automated identity verification. Recent advances in artificial intelligence and computer vision have enabled the development of robust methods for automated human presence detection and identity recognition. Because the human face constitutes a unique biometric trait exhibiting high inter-subject variability, face-based authentication is inherently resistant to impersonation attacks. Crucially, facial recognition enables accurate subject identification without necessitating physical contact, distinguishing it from contact-based biometric modalities such as fingerprint scanning. Facial recognition presents significant advantages for deployment in automated attendance management systems, substantially mitigating the risk of identity spoofing and proxy attendance fraud. While facial recognition technology had many advantages, its accuracy in recognizing faces, especially for attendance systems, required further training and testing. It surpassed other biometric recognition techniques such as fingerprint, palmprint, and iris recognition due to its non-contact method. Face recognition has been widely deployed across diverse application domains, encompassing social media platforms, border control and airport security, transportation hubs, and forensic criminal investigations. For instance, Meta's Facebook platform has historically employed automated facial recognition to suggest identity tags in user-uploaded photographs, representing one of the earliest large-scale deployments of the technology in consumer applications. Robust face recognition in both static images and dynamic video sequences remains an active and challenging research problem within the fields of image processing and pattern recognition. Reliable identification across all operational conditions—including illumination variation, subject aging, pose variation, and partial occlusion—demands large-scale annotated training datasets and high-dimensional discriminative feature representations. A substantial body of recent empirical research demonstrates that facial recognition systems have achieved significant gains in accuracy, scalability, and robustness under real-world deployment

conditions. The emergence of deep learning architectures has provided effective solutions to numerous problems previously intractable for classical machine learning algorithms, particularly in high-dimensional

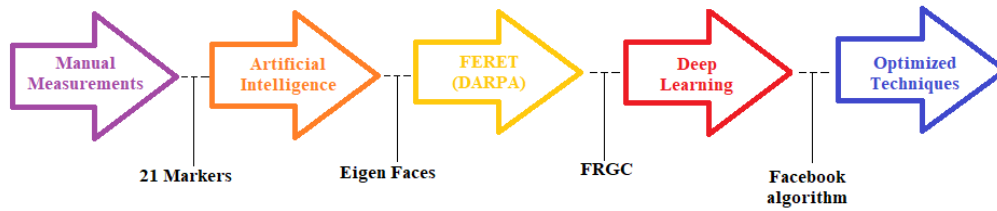


Fig 1 Development of Facial Recognition Techniques

perceptual tasks. Convolutional Neural Networks (CNNs), in particular, have fundamentally transformed the methodology for addressing computer vision and image processing tasks, achieving state-of-the-art performance across benchmarks such as face detection, object recognition, and scene classification. CNN-based facial recognition systems have gained widespread adoption owing to their demonstrated effectiveness in automated face detection, feature extraction, and identity verification. Figure 1 depicts the evolution of face recognition methods. The following subsections review the principal computational techniques that have been widely employed for facial recognition. Feature-based methods extract local facial landmarks—such as the eyes, nose, and mouth—and subsequently channel these features into a structured classifier informed by their spatial locations and local statistical descriptors, including geometric measurements and appearance-based representations. Robust feature localisation under varying illumination, pose, and occlusion conditions constitutes a fundamental challenge for feature-based facial recognition approaches.

Holistic Method: Holistic methods treat the entire face region as a unified input to the recognition pipeline, encoding global facial structure rather than isolating discrete landmarks. Representative holistic techniques include Eigenfaces [3], which apply Principal Component Analysis (PCA) for dimensionality reduction, Linear Discriminant Analysis (LDA) [4] for class-discriminative subspace projection, and Independent Component Analysis (ICA) for statistically independent feature decomposition. Holistic facial recognition systems typically comprise several sequential processing stages to match an input face against enrolled identities in a reference database. PCA is applied to the training image corpus to derive a set of Eigenfaces—orthogonal basis vectors that span the face space—enabling each face image to be compactly represented as a weighted linear combination of these basis vectors. The identity whose stored Eigenface representation yields the minimum distance to the probe image's weight vector is returned as the recognition result, enabling accurate identification across varied operational conditions.

Hybrid Method: Hybrid facial recognition systems integrate holistic and feature-based methodologies, frequently incorporating three-dimensional (3D) imaging to achieve a more comprehensive structural analysis of the face. By representing facial geometry in three dimensions, such systems are able to encode fine-grained morphological details—including orbital socket curvature and mandibular contour—even under large pose variations such as profile orientations [29]. The hybrid recognition pipeline typically comprises the following stages: (i) Face Detection, in which faces are localised through structured-light scanning or real-time video acquisition; (ii) Pose Estimation, which determines the head's spatial location, scale, and orientation within the scene; (iii) Feature Metric Assignment, in which quantitative measurements are computed for salient facial surface regions, including periocular contours and nasal angular profiles; (iv) Template Encoding, which transforms the extracted measurements into a compact numerical descriptor; and (v) Identity Matching, in which the encoded descriptor is compared against stored facial templates in the reference database to determine identity.

2 Related Work

Facial recognition algorithms are broadly classified into two principal categories: those that autonomously identify and select optimal facial features, and those that rely on manual, expert-guided feature selection [6]. Shrestha and Panday contend that robust facial recognition performance is contingent upon accurate feature extraction and classification [5]. Facial recognition has been deployed across diverse application domains, including automated student attendance monitoring in educational settings [7, 8]. Whereas human observers perceive and identify faces with remarkable speed and accuracy, replicating this capability

computationally has historically presented substantial challenges. The human visual system engages complex hierarchical neural pathways to process static and temporally dynamic facial attributes, facilitating both identity recognition and contextual interpretation of faces.

CNN fundamentally transformed methodological approaches to computer vision and image processing tasks by achieving state-of-the-art performance across a range of benchmark datasets. CNN-based facial recognition systems grew in popularity due to their effectiveness in ensuring authenticity by automatically detecting faces. Biometric authentication systems leverage a spectrum of physiological and behavioral characteristics, encompassing static modalities such as fingerprints, iris patterns, hand geometry, and DNA, as well as dynamic modalities including voice, gait analysis, and palm and finger vein patterns. Jayapriya et al. have provided a systematic review of biometric template protection techniques, specifically feature transformation-based approaches (also referred to as cancellable biometrics) and bio-cryptosystem schemes [28]. Hybrid template protection schemes are constructed by integrating feature transformation and bio-cryptosystem techniques to enhance the security of multimodal biometric templates. The work presented in [28] advances scholarly understanding of template protection methodologies and their application within multimodal biometric systems for user authentication.

Deep learning (DL) techniques have been widely applied across diverse domains, including social media analytics, disease early detection in clinical healthcare systems, and collaborative filtering-based recommendation systems [30–33]. Bruce and Young proposed an influential cognitive model positing that face perception involves the coordinated operation of multiple functionally independent processing stages. These processing stages extract facial attributes—including perceived age, gender, and affective expression, from low-level physical facial features, collectively forming a person-specific face recognition unit. This representational structure enables robust face recognition across varied and challenging perceptual conditions, including changes in viewpoint, illumination, and partial occlusion. Several empirical studies have systematically investigated the relative diagnostic salience of individual facial regions—including the eyes, mouth, and nose—for accurate face recognition. The history of face recognition technology extends to the 1950s and 1960s, with systematic automated recognition research gaining momentum during the 1970s. Early computational methodologies relied on geometric feature descriptors derived from Euclidean distances between anthropometric facial landmarks. The 1990s witnessed a substantial expansion in face recognition research, driven by concurrent advances in computational hardware capabilities and a growing demand for automated security and surveillance applications. An early semi-automated face recognition system attributed to Woodrow Wilson Bledsoe (1966) employed a manually curated facial database to identify correspondences between probe facial images and stored gallery entries. In academic environments, the deployment of facial recognition technology has been reported to reduce administrative time overhead for educators and learners while simultaneously mitigating fraudulent or proxy attendance [11]. According to Mansoor et al. [10], The system presented in [9] comprised a pipeline in which student video streams were captured, decomposed into discrete frames, and cross-referenced against a stored facial database to determine individual attendance status

Obaida et al. conducted a systematic comparative evaluation of the Viola–Jones and YOLOv3 face detection algorithms [12]. It systematically covered the procedures involved in face identification using Viola–Jones, as well as a full algorithmic description of YOLOv3 and its design. The experimental results revealed that the Viola–Jones detector achieved a detection accuracy of 45%, reflecting its prioritization of computational speed, whereas YOLOv3 attained a substantially higher detection accuracy of 90%. The comparative analysis elucidated the speed–accuracy trade-off inherent in both approaches, motivating exploration of algorithmic refinements and potential hybrid architectures designed to leverage the complementary strengths of each method. This research was consistent with the larger goal of building a face identification system that balanced speed and precision for practical applications. Kamgar-Parsi et al. addressed the problem of reliably identifying and tracking individuals listed in a surveillance watchlist [13]. To improve the recognition system's precision, the study introduced novel methodological contributions, including image morphing and automated training exemplar generation with associated ground-truth labelling, to enhance recognition precision. A critical step was the customized morphed training sets facilitated optimization of the recognition model's decision boundaries. The classification criterion was defined by whether test image feature vectors fell within the learned decision region of the network, with the scalar output of the corresponding output neuron serving as the classification score. The system classified a test image as a match when the activation of the designated output neuron exceeded a threshold of 0.5, demonstrating the reliability and computational efficiency of the proposed approach in identifying enrolled watchlist individuals. Garg et al. proposed a robust liveness detection framework for facial recognition incorporating three complementary methods: LBP-SVM, LBP-DBN, and DeBNet [14]. The proposed framework [14] employed Deep Belief Networks (DBNs) for hierarchical feature extraction and spoof classification, adopting an integrated end-to-end strategy to enhance cross-dataset

recognition accuracy. The work not only addressed the challenge of improving anti-spoofing performance across both 2D and 3D face spoof datasets, but also systematically evaluated cross-dataset generalization scenarios, underscoring the approach's applicability to real-world deployment conditions. The framework maintains the reliability and classification accuracy of the face recognition system under adversarial spoof conditions through the integration of complementary machine learning methodologies. Budiman et al. developed a university student attendance management system that replaced legacy radio-frequency identification (RFID)-based tracking with a vision-based facial recognition mechanism [16]. The study benchmarked multiple facial recognition algorithms, including Convolutional Neural Networks (CNNs), Local Binary Pattern Histogram (LBPH), Eigenfaces, Principal Component Analysis (PCA), and Haar Cascade — to determine which approach yielded superior recognition accuracy in an automated attendance monitoring context. Specifically, LBPH achieved a recognition accuracy of 77.55%, Eigenfaces 85%, PCA 80%, Haar Cascade 92.9%, and CNN attained the highest accuracy of 95%. The results demonstrated that CNN-based facial recognition yielded superior performance across diverse operational environments encompassing varied illumination, pose, and background conditions. The proposed methodology [16] enabled the institution to achieve measurable improvements in both attendance recognition accuracy and administrative processing efficiency.

Bhoyar et al. presented an enhanced face detection and real-time face recognition framework that integrated the Viola-Jones detector (implemented via OpenCV) with the LBPH descriptor and employed Support Vector Machines (SVMs) for feature classification [15]. Even under challenging operational conditions, including variable illumination, background clutter, multiple concurrent backgrounds, and partial facial occlusion the integrated algorithm achieved recognition accuracy in the range of 85–95%. Quantitative performance evaluations, incorporating deployment-phase recognition rate metrics and subject-count analyses, demonstrated the LBPH algorithm's superior recognition performance relative to the alternative methods assessed. Overall, Bhoyar et al. demonstrated that the proposed system constitutes a robust solution for face detection and identification that is both highly accurate and applicable across diverse real-world operational settings [15]. Menon et al. proposed a comparative evaluation of region-based convolutional neural network architectures, specifically R-CNN and Fast R-CNN, with the objective of improving face detection processing throughput [17]. Additionally, the YOLOv3 algorithm was evaluated as a candidate architecture for improving real-time detection throughput. The experimental implementation included a direct performance comparison between the YOLOv3 and R-CNN architectures with respect to face detection accuracy and inference speed. The results indicated that while YOLOv3 achieved superior inference speed, its face detection accuracy was lower than that of established region-based detection methods such as R-CNN. Hari et al. developed a CNN-based facial recognition system and benchmarked its recognition performance against established baseline approaches [18]. The methodology encompassed dataset collection, data pre-processing, and CNN model training to perform face detection, geometric alignment, and identity recognition in real time by matching extracted facial feature representations against a stored gallery database. The results showed that the CNN-based strategy outperformed traditional methods in terms of accuracy and efficiency. This work highlighted the benefits of CNN-based facial recognition systems and promoted continuous improvement through large-scale data training in order to improve prediction. The proposed method by Gowda et al. comprehended the prior efforts in developing student attendance systems by employing face recognition technology [27]. This study undertook a review of existing literature pertaining to student attendance management systems using facial recognition. It provided detailed insights into the methodologies, strategies, and obstacles associated with the application of face recognition techniques. Research focusing on face recognition acknowledged a number of difficulties. In response to these challenges, this study proposed a camera-based student attendance system as face recognition technology.

Koodalsamy et al. built a face recognition system using Python's OpenCV module and deep learning techniques [19]. The approach used Haar cascade methods to recognize faces and facial features were retrieved using a combination of CNN methods and the LBPH algorithm. It had a useful attendance tracking tool that marked a face as present based on authentic check-in and check-out timings; otherwise, the face was tagged as unknown. The implementation was efficiently built in Python and thoroughly evaluated on the Raspberry Pi 3 B, providing successful real-time face detection and recognition inside video frames. Notably, the study demonstrated the efficacy of integrating ResNet, SVM, and Haar Cascade approaches to improve the accuracy of face recognition systems. The work done in [19] provided a comprehensive and practical solution for effective facial recognition and attendance management similar to the system proposed by Kumar et al. [26]. It comprised four stages: database creation, face detection, face recognition, and attendance updates. Initially, databases were populated with images of students attending classes. The Haar-Cascade classifier and the Local Binary Pattern Histogram method were employed for face detection and recognition, respectively. This enabled the system to identify and distinguish faces in real-time video feeds from classrooms.

Chen et al. proposed an iris and facial biometric classifier that integrates wavelet-based probabilistic neural networks for multimodal recognition [20]. One-dimensional (1-D) wavelet transform coefficients and 1-D energy signals were extracted as discriminative features from both the iris and facial regions. The matching scores produced by a phase-information-based iris verifier and a Laplacian-eigenmap face verifier were fused at the score level using a support vector machine (SVM) fusion rule. This approach was designed to enhance the accuracy and robustness of multimodal face-iris recognition by leveraging wavelet decomposition and probabilistic neural network classifiers. Zhang et al. addressed the challenge of fusing face and iris biometric modalities captured under near-infrared (NIR) illumination using a single acquisition sensor [21]. Frontal facial images were acquired using a 10-megapixel charge-coupled device (CCD) camera. Local Binary Pattern (LBP) histogram matching, as described by Li et al. [22], was employed for eye region detection and geometric face alignment. Face recognition was performed using the Eigenface method, whereas iris recognition relied on Daugman's Gabor-filter-based phase-encoding algorithm. Score-level fusion was performed by applying sum and product combination rules following min-max normalization of the individual matcher outputs. This framework provided a cohesive methodology for fusing face and iris biometric modalities under constrained NIR illumination, demonstrating that modality-specific recognition algorithms can be effectively integrated at the score level.

Komlavi et al. conducted a comparative study of facial recognition techniques, systematically evaluating classical machine learning and deep learning classification methodologies [23]. The study benchmarked seven classification algorithms—support vector machines (SVM), artificial neural networks (ANN), k-nearest neighbours (KNN), random forests (RF), logistic regression (LR), naïve Bayes classifiers (NB), and convolutional neural networks (CNN)—across multiple facial recognition datasets. Performance was evaluated using recognition rate, confusion matrix analysis, and the area under the receiver operating characteristic (ROC) curve (AUC). The results demonstrated that CNN-based classifiers consistently outperformed all other evaluated algorithms, particularly on the ORL face database. Among classical machine learning approaches, ANN classifiers yielded competitive performance on the Yale face database. The study discussed the superior effectiveness of deep learning architectures and recommended data augmentation as a strategy to mitigate class imbalance and overfitting. The study concluded that CNN achieved the highest classification accuracy for facial recognition, surpassing all other evaluated methods, with reported recognition rates ranging from 94.82% to 98.95%. Rao [25] proposed AttenFace, an automated face-recognition-based attendance management system designed to replace the labour-intensive, error-prone process of manual attendance recording in educational institutions. Leveraging real-time face recognition, AttenFace was designed to automate attendance tracking by continuously detecting, identifying, and recording student presence. By capturing snapshots of classrooms through live camera feeds, the system identified students and recorded their attendance based on multiple frames sampled from live classroom camera feeds throughout the class session. This architecture supported independent, parallelised face recognition pipelines for each classroom, thereby ensuring horizontal scalability proportional to the number of concurrently active sessions. Furthermore, the architectural decoupling of the face recognition server from the back-end attendance calculation server facilitated seamless interoperability with existing learning management systems (LMS) such as Moodle. The face recognition algorithm operated on sampled classroom frames at 10-minute intervals, substantially reducing computational overhead relative to continuous live-feed processing. Additionally, AttenFace incorporated a threshold-based attendance policy that tolerated brief absences from the classroom, granting attendance credit to students whose presence was confirmed in at least a specified minimum number of sampled frames. Beldi and Makrem [24] investigated human face detection and recognition using machine learning algorithms for general-purpose identity verification applications. The aim was to develop an accessible Python-based framework for face recognition, promoting innovation in the field. Using TensorFlow and Python, the proposed approach fine-tuned a CNN model trained on the AT&T face dataset, achieving state-of-the-art recognition accuracy. Model performance was further improved through systematic addition of convolutional and pooling layers to the network architecture. The study acknowledged the critical role of recognition accuracy and emphasised the practical importance of minimising inference latency for real-world deployment. The study further explored the trade-offs among recognition accuracy, computational efficiency, and memory resource allocation inherent in the practical deployment of face recognition systems.

3. The Proposed DL-based Secure Real-Time Face Recognition System

In the domain of Android GUI development using the PyQt designer tool, the paper focused on crafting a user-friendly application tailored for facial image capture. The proposed system acquires facial images seamlessly by incorporating fluid camera functionality and contextual user prompts designed to maximise user engagement. PyQt deploy is a tool included with Qt that facilitates the deployment of

applications developed using the PyQt framework in Python to the Android platform. The face acquisition module, tightly coupled with the device camera API, enables users to capture facial images directly within the Android application. Image quality optimisation techniques—including real-time luminance feedback and facial alignment guidance—are incorporated into the interface to improve the overall usability of the capture workflow. Real-time face detection is accomplished by integrating the Viola-Jones algorithm, which identifies and spatially localises facial regions within each captured image frame. The flowchart of the proposed face recognition system was illustrated in Figure 3. The facial encryption module employs the Chaotic Logistic Map to guarantee confidentiality and integrity throughout both the encryption and decryption phases. The module transforms pixel intensity values within the acquired facial images through chaotic permutation and diffusion operations, thereby achieving robust encryption while preserving discriminative biometric features. Pre-processed facial images drawn from a labelled dataset persisted in a PostgreSQL relational database are used as training data for the CNN model. This integrated dual-module approach addresses the complementary requirements of cryptographically secure facial data protection and high-accuracy CNN-based facial recognition, collectively advancing the system's security posture and recognition performance. The backend infrastructure was implemented using PostgreSQL as the relational database management system (RDBMS), providing a structured and persistent foundation for storing critical system data. The database schema was carefully engineered to accommodate user profile records and corresponding attendance logs, thereby enforcing a normalized and well-structured data model. In the recognition and verification phase, the inference logic for CNN-based facial recognition was integrated into the system pipeline. The identity verification process entailed matching the CNN-generated facial embeddings against stored biometric records in the database to validate recognition accuracy and system reliability. Usability was addressed by embedding real-time user feedback mechanisms within both the facial image acquisition and recognition stages of the interaction workflow. The graphical user interface (GUI) was designed to present recognition outcomes and contextual metadata in real time, thereby supporting responsive and informative human–computer interaction. Continuous updates to the database records ensured the system remained accurate and up to date with the latest attendance results.

Live facial images captured via camera sensors constitute the primary input data source for the recognition model. Each captured frame is persisted to storage upon completion of a predefined number of acquisition iterations. The collected image dataset is subsequently subjected to preprocessing and used for model training. Each input image has spatial dimensions of 250×250 pixels. Each facial sample is initially acquired as a 250×250 pixel image cropped and centred on the region of interest corresponding to facial features. The acquired facial images are subsequently encrypted using a Chaotic Logistic Map algorithm, producing visually randomized grayscale representations that render the biometric data unintelligible to unauthorized parties, thereby enhancing data confidentiality. The CNN model and the Viola–Jones face detection algorithm were evaluated across multiple train–test split ratios; the 90:10 partition yielded the highest classification performance with respect to accuracy, precision, and F1-score. Figures 2.1 through 2.5 present the confusion matrices corresponding to each evaluated algorithm.

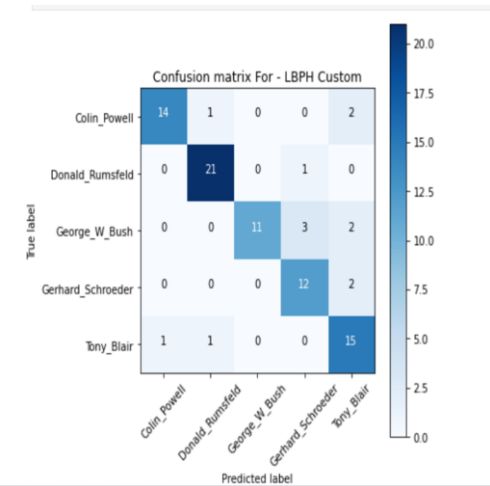


Fig. 2.1 Confusion Matrix of LBPH Algorithm

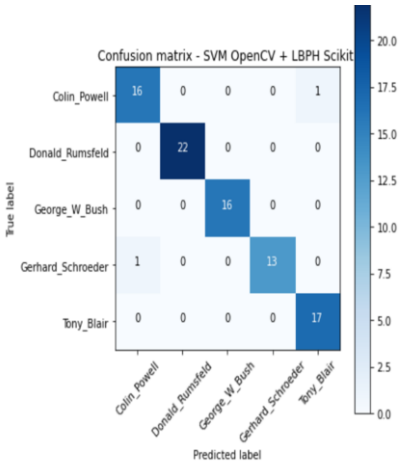


Fig. 2.2 Confusion Matrix of SVM, OpenCV and LBPH Algorithm

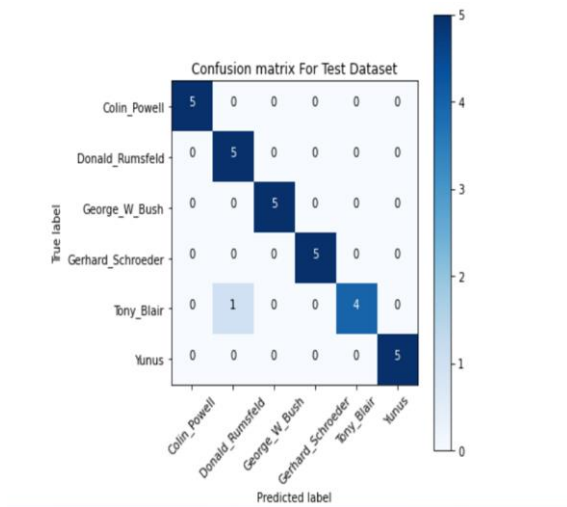


Fig 2.3 Confusion Matrix of Eigen Face Algorithm

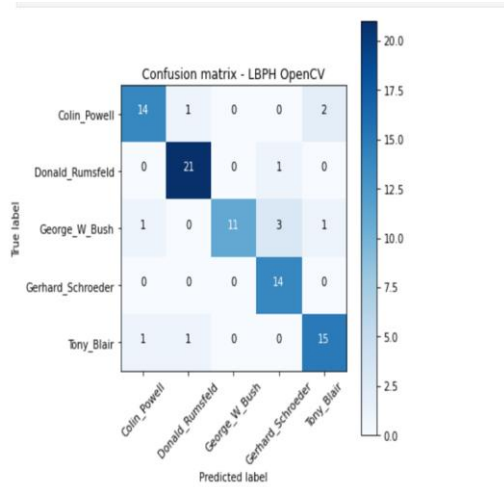


Fig 2.4 Confusion Matrix of OpenCV

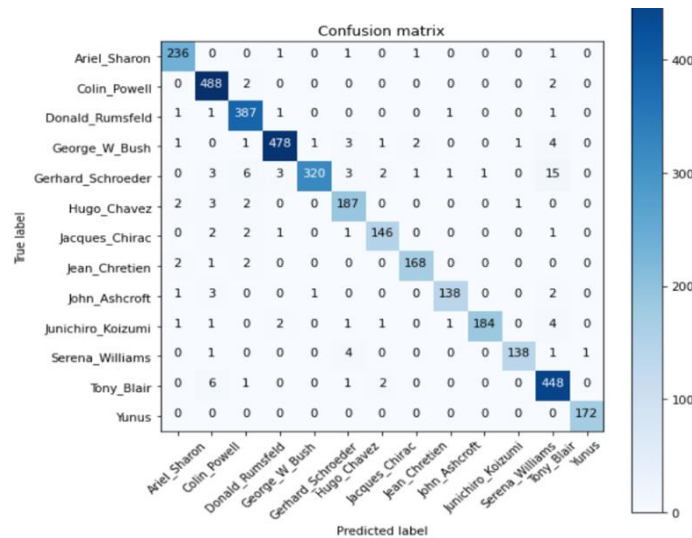


Fig 2.5 Confusion Matrix of the Proposed Convolutional Neural Network Classifier

3.1 Viola Jones Detection

The Viola–Jones object detection framework, proposed by Viola and Jones [ref] in 2001, represented a seminal advancement in computer vision by achieving, for the first time, real-time object detection at competitive accuracy rates. Although the framework is capable of detecting objects in real time, its predominant application in the literature has been face detection, leveraging Haar-like features and an integral image representation to accelerate the feature computation process. The framework achieves a high detection rate, characterised by an elevated true-positive rate and a correspondingly low false-positive rate, owing to its cascaded AdaBoost classifier architecture, which progressively rejects non-object sub-windows and thereby enables efficient, high-throughput image processing. The primary objective of this stage was face detection, as distinguished from face recognition, wherein the system discriminates between facial and non-facial image regions as a prerequisite processing step. The Viola-Jones algorithm comprises several interrelated modules, each of which is described in the subsections that follow.

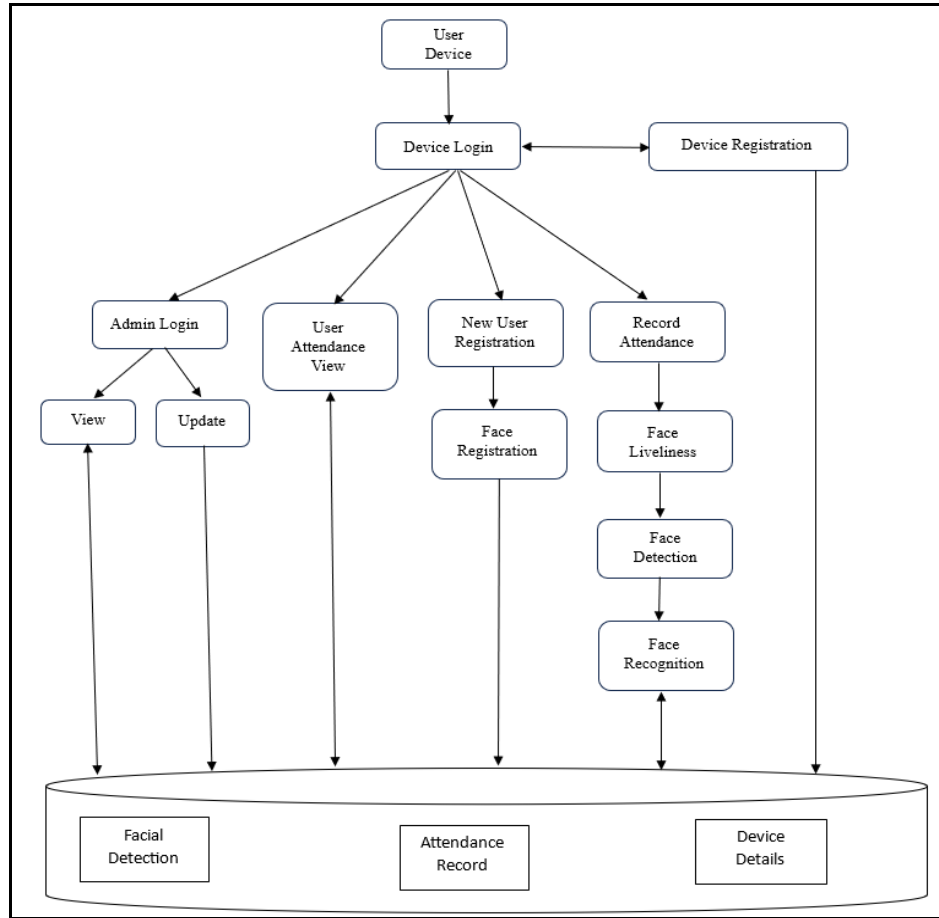


Figure 3 System Flowchart of the Proposed Deep Learning-Based Secure Real-Time Face Recognition Architecture

Haar-like Features: This module employs rectangular filter patterns, commonly referred to as Haar-like features, to encode salient structural attributes of image regions, including edges, corners, and linear gradients. Canonical human facial structure exhibits consistent photometric properties, such as higher pixel intensity in the upper cheek regions relative to the periorbital area, and greater luminance along the nasal bridge compared with the eye sockets. These photometric regularities reflect the spatial configuration of discriminative facial landmarks, including the relative positions and dimensions of the eyes, mouth, and nasal bridge. Pixel intensity gradients encode directional magnitude information that underlies the discriminative power of Haar-like features. Each input image is subdivided into a sliding detection window of 24×24 pixels, which serves as the base resolution for feature computation. Each Haar-like feature is subsequently applied across the sub-regions of the detection window to compute feature response values. Each Haar-like feature is anchored to a specific spatial location, scale, and orientation within the detection sub-window.

Several computational and representational advantages motivate the use of aggregate region features over raw per-pixel values in this framework. The Haar-like feature-based approach demonstrates substantially superior computational performance relative to direct pixel-based classification methods, owing to reduced feature dimensionality and faster evaluation via the integral image representation. The rectangular filter responses employed in this framework are structurally analogous to Haar wavelet basis functions, which are well established in signal processing for multi-resolution image analysis rather than direct pixel-level classification. The Haar-like feature set in the Viola-Jones framework encompasses three canonical feature types: two-rectangle, three-rectangle, and four-rectangle features. The response of a two-rectangle feature is computed as the absolute difference between the summed pixel intensities within two adjacent rectangular regions of equal dimensions. The two constituent rectangular regions are identical in size and shape, and are arranged in either a horizontal or vertical adjacency configuration. Figure 4 illustrates the spatial arrangement and computational operation of representative Haar-like rectangular feature types. The response of a three-rectangle feature is computed as the difference between the combined pixel-intensity sum of the two outer rectangular regions and the pixel-intensity sum of the central rectangular region. The four-rectangle feature

computes the difference between the pixel-intensity sums of two diagonally opposite pairs of rectangular regions, enabling the detection of diagonal edge and gradient patterns.

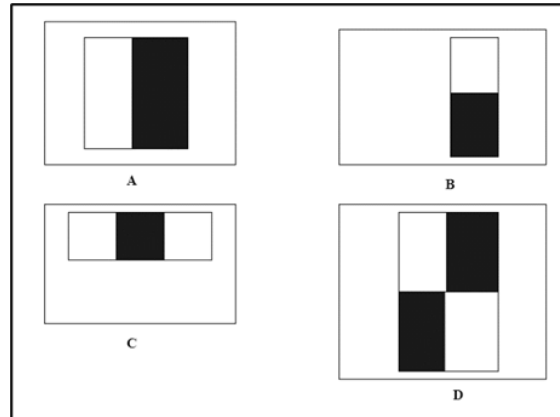


Figure 4. Representation of a Haar-Like Features

Integral Image To accelerate feature computation, the Viola-Jones algorithm converts the input image into an integral image representation, also known as a summed-area table. Rectangular Haar-like feature responses are computed with $O(1)$ time complexity per feature evaluation through the intermediary integral image representation, regardless of the size of the rectangular region. At any coordinate location (x, y) in the image domain, the integral image value $ii(x, y)$ is defined as the cumulative sum of all pixel intensities at positions (x', y') such that $x' \leq x$ and $y' \leq y$. This integral image representation enables constant-time computation of arbitrary rectangular region sums, thereby substantially reducing the computational cost of evaluating the full Haar-like feature set. Formally, the integral image $ii(x, y)$ accumulates the pixel intensities of the original image $f(x', y')$ over the inclusive rectangular region bounded by the origin and the point (x, y) , such that $ii(x, y) = \sum f(x', y')$ for all $x' \leq x$ and $y' \leq y$.

$$ii(x, y) = \sum_{\substack{x' \leq x \\ y' \leq y}} I(x', y') \quad \text{-----(1)}$$

$$ii(x, y) = I(x, y) + ii(x, y - 1) + ii(x - 1, y) - ii(x - 1, y - 1) \quad \text{----(2)}$$

Adaboost Learning: The AdaBoost (Adaptive Boosting) algorithm iteratively selects and combines a sequence of weak binary classifiers, each performing marginally better than chance, to construct a strong composite classifier through a weighted linear ensemble. A 24×24 detection window yields approximately 160,000 possible Haar-like feature instances when all valid rectangular sub-region combinations are enumerated. From this pool of over 160,000 candidate features, AdaBoost assigns discriminative weights to a small subset of the most informative features, producing a weighted strong classifier. The weighted combination of the selected discriminative features collectively determines whether a given detection window contains a face, constituting the binary classification decision. In the AdaBoost framework, each individual selected feature, combined with its associated threshold, constitutes a weak classifier—a binary decision function that achieves accuracy only marginally above random chance. The AdaBoost selection procedure identifies a compact, maximally discriminative subset of features from the full candidate feature space, significantly reducing the computational burden of subsequent classification.

Cascading Classifiers: The attentional cascade, also referred to as the cascade classifier, constitutes the final and most computationally significant module of the Viola-Jones detection framework. The cascade classifier is architecturally organised as a sequential series of stages, each comprising an increasingly complex strong classifier trained on progressively harder negative examples. The cascade architecture achieves high detection throughput by applying an early-rejection strategy: sub-windows classified as non-face by an early-stage classifier are immediately discarded, precluding evaluation by subsequent, more complex stages. Consequently, the vast majority of computational resources are concentrated on the small fraction of sub-windows that pass all cascade stages, enabling real-time face detection performance. Haar's rectangular characteristics were applied to the face to determine the difference. The cascade classifier is applied to a grayscale image representation,

within which it evaluates the intensity distribution across rectangular sub-regions rather than performing pixel-level binary thresholding. Many classifiers were pre-trained based on face and eyes, which were already available in OpenCV and XML files. The pseudo-code below illustrated how the algorithm helped in face detection with liveness.

3.2 Convolutional Neural Network

A CNN receives image data as input and extracts discriminative features or objects that are subsequently utilized by classification algorithms to distinguish one image from another [23]. The architecture of a CNN is inspired by the hierarchical organization of synaptic connectivity observed in the mammalian visual cortex. The architecture of a CNN comprises the following principal components:

Convolutional Layer: Convolutional layers apply learned filter kernels to detect spatial patterns, edges, and hierarchical features within the input data. By sliding convolutional filters over the input, relevant features are captured by systematically sliding learnable convolutional filters across the input volume, thereby generating corresponding feature maps. The convolution operation employs zero-padding at the spatial boundaries of the input tensor to preserve its dimensions and ensure well-defined border computations. The purpose of zero-padding is to ensure numerically valid outputs at spatial boundaries, wherein corresponding elements of the input tensor and the convolutional kernel are element-wise multiplied and accumulated via summation. As illustrated in Figure 6, the convolution kernel undergoes a 180-degree rotation about its centre, which is mathematically equivalent to successive horizontal and vertical reflections of the kernel matrix.

Pooling Layer: Pooling layers, also referred to as subsampling or downsampling layers, reduce the spatial dimensions of the feature maps produced by preceding convolutional layers, thereby decreasing computational cost and controlling overfitting. Widely adopted pooling strategies include max pooling, which retains the maximum activation within each pooling window, and average pooling, which computes the mean activation across the window, as illustrated in Figure 7.

Fully Connected Layers: Situated in the terminal stages of the network, fully connected layers establish dense connections between every neuron in one layer and every neuron in the subsequent layer, forming a complete bipartite graph between adjacent layers. This dense connectivity enables the network to integrate globally distributed feature representations, supporting the learning of complex, non-linear decision boundaries required for multi-class classification tasks.

Activation Layer: Activation functions introduce non-linearities into the network's transformation pipeline, enabling the model to approximate complex, non-linear mappings between input features and target outputs that would be unachievable with purely linear operations. Commonly employed activation functions include the Rectified Linear Unit (ReLU), the logistic sigmoid function, and the hyperbolic tangent (tanh) function, each exhibiting distinct properties with respect to gradient flow and saturation behaviour.

Batch Normalization: This module standardizes the inputs of each layer by re-centering and rescaling the activations prior to passing them to the subsequent layer. By improving convergence and stability, batch normalization reduces internal covariate shift, thereby accelerating convergence and improving training stability in deep neural architectures.

Algorithm 1 Viola-Jones Face Detection Algorithm

Input: Original test image,
Output: Image with face indicated in frame
begin:
for i=1 to num of scales in pyramid of images do
{
Downsample image to create images
Compute integral image
for j=1 to num of shift steps of sub-window do
{
for k= -1 to num of filters of stages in cascade classifier do
{
for l=1 to num of filters of stage k do Filter detection sub-window
{
Accumulate filter outputs
}
if accumulation fails per stage threshold
{
Reject sub-window as face Break this k for loop
}
}
if sub-window passed all per-stage checks
{
Accept this sub-window as a face
}
}
}
}
end

Figure 5. Pseudocode illustrating the operational pipeline of the Viola–Jones object detection algorithm.

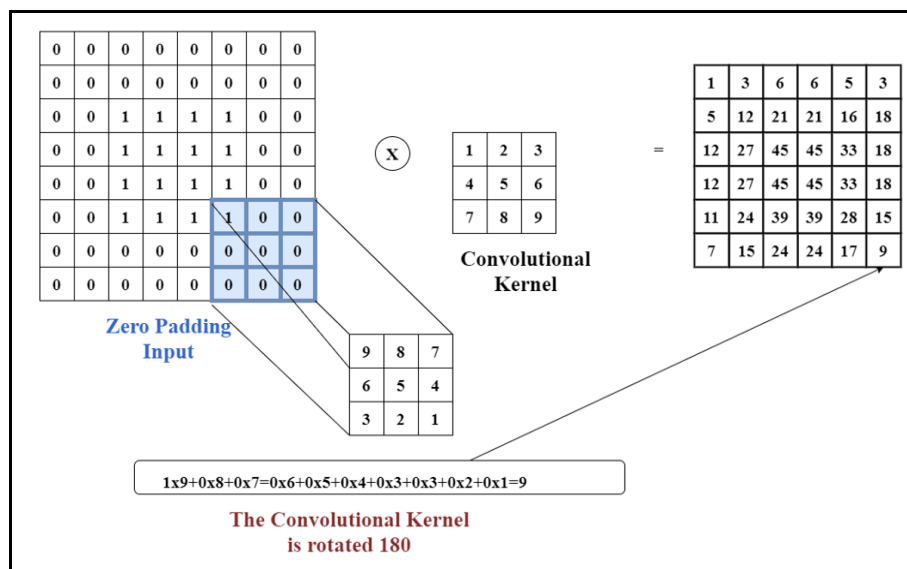


Figure 6 Implementation of Convolutional Layer

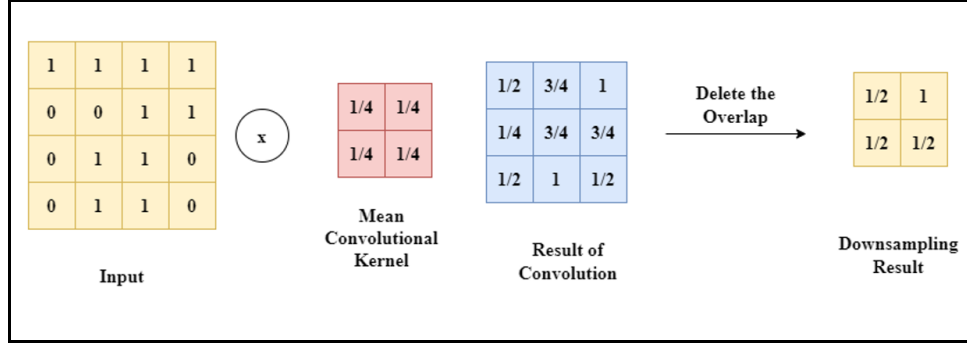


Figure 7 Pooling Operation

$$\text{ReLU} : f(x) = \max(0, x). \quad \text{----- (3)}$$

$$\text{sigmoid} : f(x) \quad \text{----- (4)}$$

$$\text{tanh} : f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad \text{----- (5)}$$

Dropout: Dropout is a regularisation technique that stochastically sets a fraction of neuron activations to zero during each forward pass in the training phase. This mechanism mitigates overfitting by preventing co-adaptation of neurons, thereby promoting improved generalisation to previously unseen data distributions.

CNN Model for Face Recognition: This study presents a custom CNN architecture designed to maximise face recognition classification accuracy. Although the proposed architecture shares structural similarities with the classical LeNet-5 model, it differs in its input dimensionality, network width, and fully connected layer configuration. The proposed CNN architecture comprises two convolutional layers (C1 and C2) and two subsampling layers (S1 and S2), following the layer-naming convention of LeNet-5. Each normalised facial image is provided to the network via a single-channel feature map at the input layer. The first convolutional layer, C1, comprises six feature maps, each of which applies a 5×5 learnable convolutional kernel to its receptive field. The first subsampling layer, S1, produces six feature maps by applying a pooling operation to the corresponding output feature maps of C1.

Data Augmentation: The Labeled Faces in the Wild (LFW) benchmark dataset was assembled to support research into unconstrained face recognition under real-world conditions. LFW was released specifically to advance academic face verification research, rather than to serve as a benchmark for the comprehensive evaluation of commercial facial recognition systems. The dataset represents an early contribution toward providing a labelled face corpus that encompasses the diverse real-world scenarios encountered in naturalistic settings. The dataset exhibits natural variation across a broad range of factors, including pose, illumination, focus, resolution, facial expression, age, gender, ethnicity, accessories, cosmetics, occlusion, background complexity, and image quality. Although the dataset provides images in a standardised format to facilitate broad utility, the total number of samples is insufficient to adequately train a deep neural network for robust facial recognition. As illustrated in Figure 8, four data augmentation techniques—horizontal flipping, translation, scaling, and rotation—are applied to address this data scarcity problem by expanding the effective size of the training dataset.

Synthetic dataset expansion via geometric and photometric image transformations constitutes a well-established strategy for mitigating the scarcity of large-scale labeled training corpora in face recognition pipelines. By generating morphologically and photometrically diverse sample variants, the augmentation procedure increases intra-class variability within the training set, thereby reducing overfitting in the CNN classifier and yielding measurable gains in face recognition accuracy. Upon application of the augmentation pipeline detailed above, the original dataset was expanded by a factor of twenty, yielding the augmented corpus whose composition is illustrated in Figure 9.

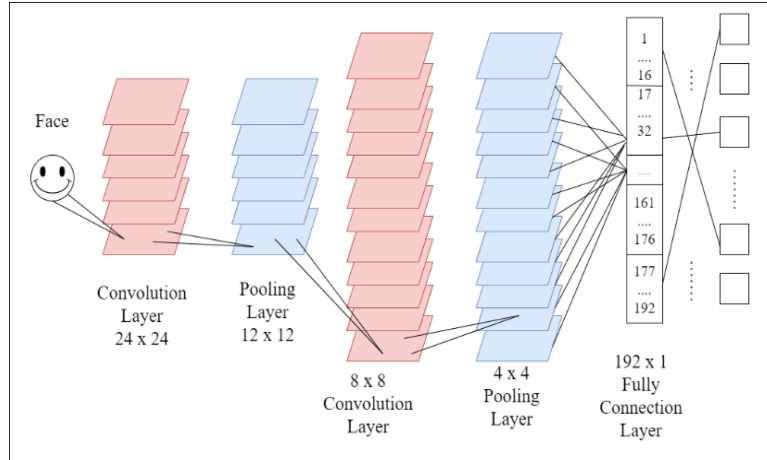


Figure 8 CNN Representation Model for Face Recognition

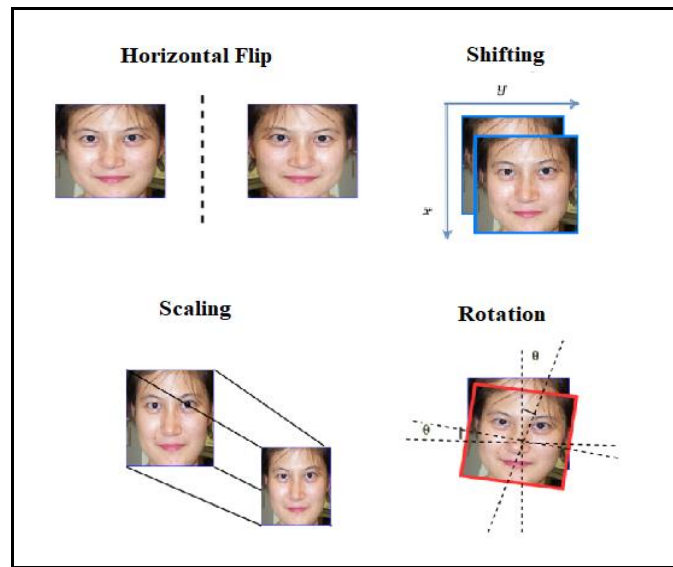


Figure 9 Methods of Augmentation

3.3 Chaotic Logistic Mapping

Within the domains of chaos theory and dynamical systems, the logistic map—a well-established nonlinear recurrence relation—occupies a central role in the study of complex iterative behaviour. For specific ranges of the control parameter r (typically $3.57 \leq r \leq 4$), the logistic map exhibits deterministic chaos, characterised by sensitive dependence on initial conditions and aperiodic long-term dynamics. The logistic map has been extensively applied across diverse disciplines, including ecological modelling of population dynamics, cryptographic key generation, and pseudo-random number generation [citation needed]. The logistic-map-based encryption scheme is adopted in this work because its chaotic pseudo-random sequences demonstrate superior unpredictability and sensitivity to initial conditions compared with symmetric algorithms such as DES and asymmetric schemes such as RSA and ECC, attributes that are critical for robust image encryption. This sensitivity to initial conditions renders the ciphertext computationally infeasible to predict or reproduce without exact knowledge of the initial seed and control parameter, thereby strengthening resistance to cryptanalytic attacks.

Pixel Representation: In a standard RGB colour image, each pixel is represented by an ordered triple of intensity values corresponding to three spectral channels: red (R), green (G), and blue (B), each quantised over the range $[0, 255]$ in an 8-bit-per-channel encoding. This additive colour model enables the specification of approximately 16.7 million distinct colours through the combinatorial mixing of the three primary channel

intensities. Concretely, a pixel is denoted as the triplet (R, G, B), where R, G, and B represent the discrete intensity values of the red, green, and blue channels, respectively. By additively combining these three channel values, the full gamut of colours reproducible within the sRGB colour space is rendered for each pixel in the image. A structured color representation is required for manipulating and processing color images across a range of applications, including image encryption, in which each color channel undergoes independent encryption operations.

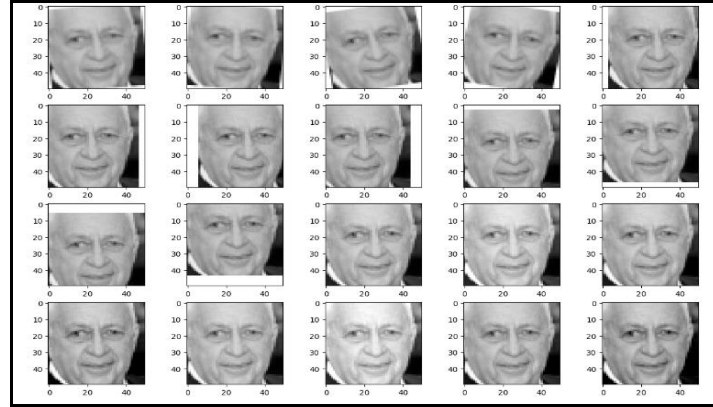


Figure 10 Representation of 20-fold Image Augmentation

Key Generation: The logistic map equation is employed to generate pseudorandom keys for each color channel of every pixel in the image. This equation generates a sequence of values exhibiting deterministic chaotic behaviour, rendering them computationally unpredictable and thus suitable for cryptographic key generation. Three distinct keys— K_r , K_g , and K_b —are generated for each pixel in the image, corresponding to the red, green, and blue channels, respectively. These keys correspond to the red, green, and blue color channels, respectively. Each key is subsequently applied to encrypt its corresponding color channel of the respective pixel. By generating a unique key for each color channel of every pixel, the encryption procedure preserves the chromatic information of the image while providing robust pixel-level security.

XOR Operation and Substitution: In the XOR operation for each color channel of a pixel, the intensity value of the channel was XORed with the corresponding key value, resulting in the encrypted intensity value. XOR Operation and Substitution: For each color channel of a given pixel, the channel intensity value is XORed with its corresponding key value, producing the encrypted channel intensity. Similarly, the green channel intensity value G is XORed with K_g to yield E_g , and the blue channel intensity value B is XORed with K_b to yield E_b . After each color channel of a pixel has been independently encrypted, the three resulting ciphertext values are combined to reconstruct the encrypted pixel.

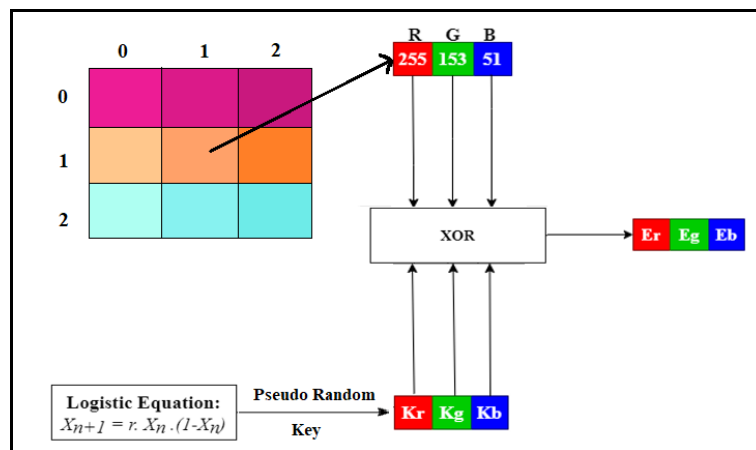


Figure 11 Representation of Chaotic Logistic Mapping Encryption

This approach yields an encrypted pixel in which each color channel is independently enciphered using its dedicated key value. Although the structural chromatic composition of the original pixel is preserved, the channel intensity values are effectively obfuscated through the XOR-based key mixing operation, as illustrated in Figure 11. This procedure is applied iteratively to every pixel in the image, ensuring that each pixel is

independently encrypted with a pixel-specific set of pseudorandom keys, after which the fully encrypted image is returned as output.

Key Retrieval: To initiate decryption, the identical set of keys employed during encryption— K_r , K_g , and K_b —must first be recovered. These keys are typically stored securely or transmitted alongside the encrypted image through an established secure channel. In the present implementation, the generated keys are persisted to a file, enabling retrieval of the original image during the decryption phase.

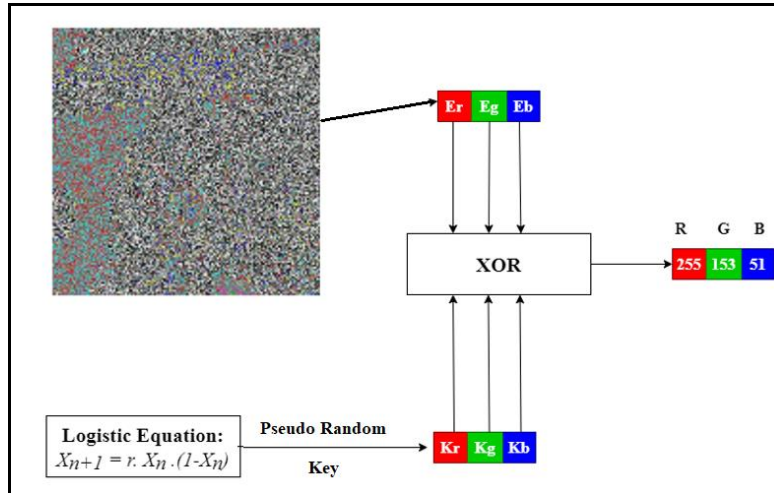


Figure 12 Representation of Chaotic Logistic Mapping Decryption

Reverse XOR Operation: The decryption algorithm inverts the XOR operation applied to each pixel of the encrypted image by employing the corresponding chaotic key values generated during the encryption phase. This operation precisely inverts the bitwise XOR transformation applied per color channel during encryption, thereby recovering the original intensity value for each channel within every pixel. Specifically, in the red channel, the ciphertext intensity value E_r is XORed with its corresponding chaotic key K_r , yielding the plaintext intensity R (i.e., $E_r \oplus K_r = R$). Analogously, the ciphertext intensity of the green channel (E_g) is XORed with the corresponding key K_g to yield the plaintext value G ($E_g \oplus K_g = G$), and the ciphertext intensity of the blue channel (E_b) is XORed with K_b to yield B ($E_b \oplus K_b = B$). Once the plaintext intensity values for all three color channels have been recovered, they are recombined to reconstruct the original pixel, as illustrated in Figure 12.

Reconstruction of Original Image: By systematically applying the inverse XOR operation—using the chaotic key values generated by the logistic map—to each pixel of the ciphertext image, the algorithm restores the original intensity levels across the red, green, and blue channels. This channel-wise inversion reconstructs the image's original chromatic data, completing the decryption of the ciphertext. The decryption process thereby yields a reconstructed image whose per-channel intensity values are identical to those of the plaintext image prior to encryption. The reconstructed image is mathematically equivalent to the original plaintext image, rendering it suitable for subsequent visual analysis, perceptual quality assessment, or further computational processing.

4 EXPERIMENTAL RESULTS

Models/Metrics	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LBPH Algorithm	85	86	84	85
SVM +LBPH+Open CV	97	98	97	98
Eigen Face	97	97	97	97
LBPH + Open CV	88	88	87	87
CNN	98	97	98	97

Table 1 Comparison results of various algorithms of Face Recognition

Table 1 presents a quantitative comparison of several face recognition approaches, including the Local Binary Pattern Histogram (LBPH), Eigenface, OpenCV-based pipelines, Support Vector Machine (SVM), Convolutional Neural Network (CNN), and their hybrid configurations. The F1-score serves as a harmonic mean of precision and recall, providing a balanced measure of classification performance, whereas overall accuracy quantifies the proportion of correctly identified subjects across all test instances. The standalone LBPH algorithm achieved an accuracy of 85%, with a precision of 86% and an F1-score of 85%, representing the lowest overall performance among all configurations evaluated. This result indicates that the standalone LBPH configuration is the least performant of all architectures evaluated, suggesting that texture-based handcrafted descriptors alone are insufficient for high-accuracy face recognition under the experimental conditions employed. The hybrid configuration integrating an SVM classifier with LBPH feature extraction and OpenCV preprocessing attained an accuracy of 97%, a precision of 98%, and an F1-score of 98%, demonstrating that combining handcrafted features with a discriminative classifier substantially improves recognition performance. The Eigenface algorithm achieved uniform scores of 97% across all four evaluation metrics—accuracy, precision, recall, and F1-score—indicating a well-balanced recognition performance, though the basis for this uniformity warrants further analysis. The hybrid pipeline combining LBPH feature extraction with OpenCV preprocessing yielded an accuracy of 88% and an F1-score of 87%, outperforming the standalone LBPH baseline but falling short of the SVM-based and CNN-based configurations. Among all configurations evaluated, the CNN-based model achieved the highest performance, attaining an accuracy of 98%, a precision of 97%, a recall of 98%, and an F1-score of 97%, consistent with the established superiority of deep convolutional architectures in learning hierarchical facial representations. So from the above table comparison, CNN can provide better predictions as compared to other models.

Data split ratio (train data: test data)	Accuracy	Precision	Recall	F1-score
60:40	90	89	89	90
70:30	92	91	92	92
80:20	95	94	94	95
90:10	98	97	98	97

Table 2 Comparison results of different test and train data split using CNN algorithm

Partitioning the dataset into distinct subsets is a foundational step in developing a face recognition system, as it enables rigorous training, evaluation, and validation of the underlying machine learning models. This partitioning strategy promotes unbiased performance estimation, mitigates overfitting, facilitates hyperparameter optimisation, and enables systematic comparison across model configurations to identify the most effective predictive approach. Table 2 presents a comparative analysis of classification performance metrics across varying train–test split ratios applied to the CNN-based model. The 60:40 train–test split yielded a classification accuracy of 90%, accompanied by precision and F1-score values of 89% and 90%, respectively. Increasing the training proportion to a 70:30 split improved accuracy, precision, and F1-score to 92%, 91%, and 92%, respectively, relative to the 60:40 baseline. These results suggest a positive monotonic relationship between the proportion of training data and both classification accuracy and F1-score, indicating that larger training sets enhance model generalisation. The comparative analysis demonstrates that the 90:10 train–test split achieved the highest classification performance, yielding an accuracy of 98% and an F1-score of 97%, thus representing the optimal partitioning configuration evaluated in this study. The deployed system employs real-time image analysis to identify registered individuals and records attendance events to the back-end database upon successful recognition. Figure 13 illustrates the graphical user interface (GUI) for attendance recording, through which users initiate the process by selecting either the check-in or check-out button. Upon button activation, the system initiates video capture and performs liveness detection to determine whether the presented face is bona fide or a spoofing artefact. Upon successful liveness verification, the facial recognition module is activated, and the corresponding attendance record is updated in the database.

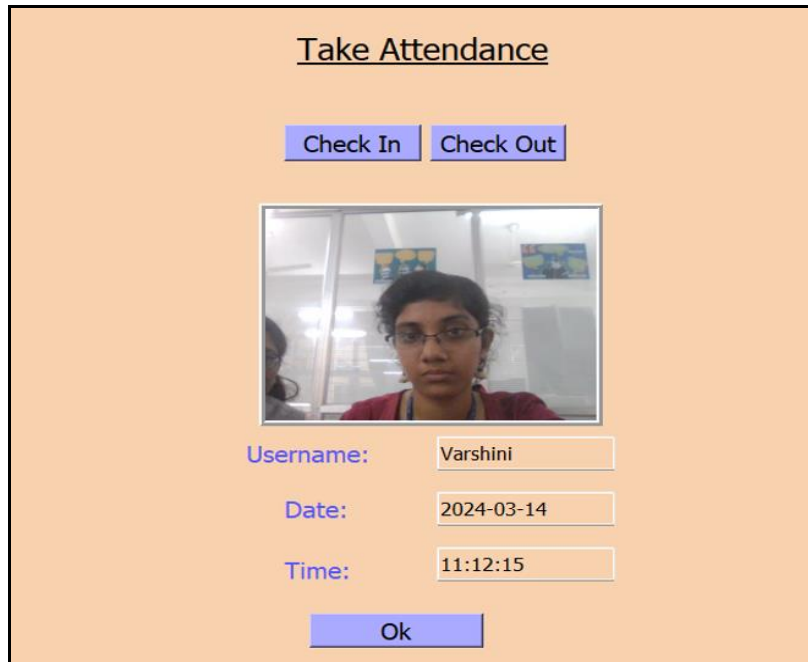


Figure 13 Attendance Recording after clicking the check in or check out button.

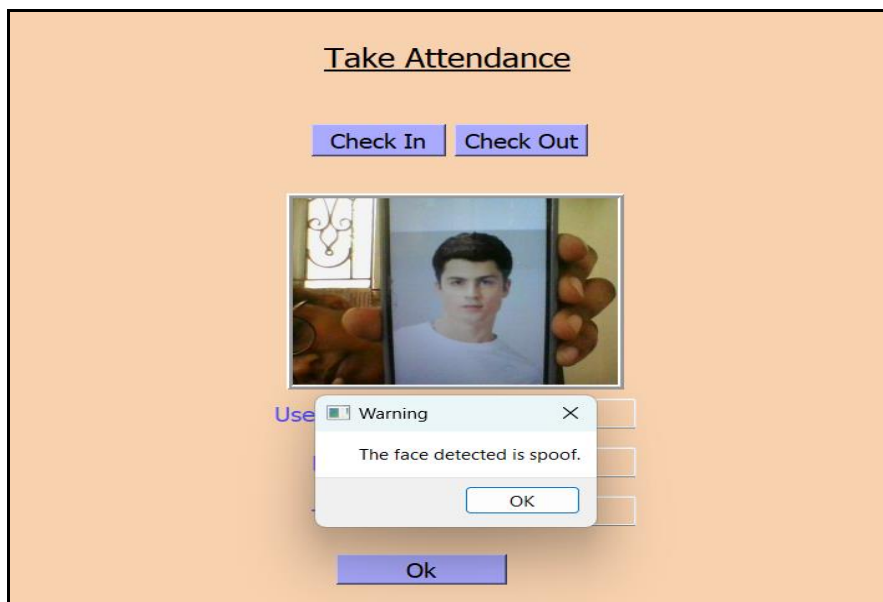


Figure 14 Experimental Detection Output for a Spoofed Facial Image Sample

Details

Name:

Varshini

User ID:

123654987

Department:

CSE

Email ID:

varshini@gmail.com

Attendance Details

No. of working days:

20

Attendance Percent:

80.0

Days Present

	Date	Check In Time
5	2024-04-03	08:19:00
6	2024-04-04	08:19:00
7	2024-04-05	08:19:00

[Back](#)

Figure 15 System Interface View Displaying Granular User Attendance Records

As illustrated in Figure 14, when the liveness detection module identifies a spoofing attempt, the recognition pipeline is immediately terminated to prevent the generation of fraudulent attendance entries, thereby preserving data accuracy and integrity. A dedicated user dashboard, depicted in Figure 15, displays personalised information including attendance percentage, total number of working days, and timestamped check-in and check-out records in a structured tabular format

4 CONCLUSIONS AND FUTURE WORK

This paper presents a computationally efficient deep learning-based automated attendance system incorporating a custom-trained convolutional neural network for simultaneous liveness detection and face recognition. Comparative evaluation demonstrates that the proposed framework achieves statistically superior precision and recall relative to established baseline methods, including the Local Binary Patterns Histogram (LBPH) classifier, Support Vector Machine (SVM)-based recognition, and the Eigenface (PCA-based) algorithm. Furthermore, the integration of a graphical user interface developed with the PyQt5 framework improved the system's usability by providing intuitive interaction mechanisms and real-time result visualisation, thereby enhancing accessibility for non-technical end users. Attendance records are persistently stored and managed using a PostgreSQL relational database management system. The proposed system demonstrates both computational efficiency and security compliance in the automated recording and monitoring of attendance data. Multiple face recognition algorithms — including Local Binary Patterns Histograms (LBPH), Eigenfaces, Support Vector Machines (SVM), Convolutional Neural Networks (CNN), and ensemble hybrid configurations were systematically evaluated using performance metrics comprising accuracy, precision, and F1-score across multiple train-test partition ratios. Empirical evaluation revealed that the CNN-based classifier achieved superior accuracy and F1-scores relative to all other evaluated models. Because the system is trained exclusively on real-world facial data and supports incremental learning upon the ingestion of new samples, its recognition performance is expected to remain robust under authentic operational conditions. Nevertheless, data quality degradation arising from sources such as motion-blurred or occluded facial captures and environmental interference may adversely affect model inference accuracy. Despite the system's demonstrated advantages, susceptibility to false positive recognition errors persists as a consequence of sensitivity to extrinsic environmental variables, including sensor resolution, illumination intensity, and ambient lighting conditions.

Future work should explore performance optimisation of the CNN-based classifier for multiclass recognition tasks and domain-adapted fine-tuned models, targeting reduced inference latency and improved computational resource efficiency without degradation of recognition accuracy. The architectural integration of cloud-based infrastructure to support distributed model retraining on newly enrolled users and scalable storage of facial embedding registries constitutes a substantive open research problem warranting systematic investigation. Additionally, continued research into algorithm comparison, deployment, and scalability aspects was crucial for ensuring the system's viability and scalability in real-world deployment scenarios.

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